

Credit card churn prediction model using Logistic Regression and Decision Tree																																																																																																																																									
Guangli Nie, Wei Rowe, Lingling Zhang, Yingjie Tian, Yong Shi																																																																																																																																									
Introduction		Modelling and Results																																																																																																																																							
Motivation To predict churn of credit card users Research problem With rising cost and extensive competition from credit card companies, a study is required to understand credit card users’ churning intention and explain such behavior traits. Data understanding Data collection process Data Source China commercial bank Data preparation process Independent variables (X1X2...) data calculated from observation period Dependent variable (Y) data calculated from performance period Observation Period T1 12 months timeframe Jan 1 2006 to Dec 31 2006 T2 T3 Performance Period T2 12 months timeframe Jan 1 2007 to Dec 31 2007 T3 Customer A is churner (No longer a customer possibly due to switch credit card company) Credit card customer CHURN Y = 1 Customer B is retained (Remained as a customer) Credit card customer Y = 0 Takeaway point Observe behaviors of customers during 1 year observation period, predict if credit card customer becomes a churner or not during 1 year performance period Balanced data mart samples Training set: Churners, 215, 4% Training set: Non Churners, 244, 4% Testing set, 4997, 92% Training set: Churners Training set: Non Churners Testing set Stepwise variables and model selection process Model 1 Customer basic information Poor data quality Model 2 Basic card information Model 3 Customer information Basic card information Higher Type I error Lower average error and misclassification cost Model 4 Transactions information Higher misclassification cost Model 5 Customer personal information Basic card information Risk related information Model 6 Customer information Basic card information Risk related information Transaction information BEST MODEL MODEL 6 2 customer personal information 4 card basic information 3 risk information 6 transaction information Higher accuracy and lower misclassification cost		Churn prediction model Goodness of fit <table><tr><th>Goodness of fit</th><th>Model 1</th><th>Model 2</th><th>Model 3</th><th>Model 4</th><th>Model 5</th><th>Model 6</th></tr><tr><td>AIC</td><td>629.15</td><td>394.20</td><td>376.62</td><td>338.50</td><td>304.12</td><td>208.15</td></tr><tr><td>SC</td><td>641.54</td><td>427.23</td><td>417.91</td><td>363.28</td><td>361.92</td><td>286.61</td></tr><tr><td>-2 Log L</td><td>623.15</td><td>378.20</td><td>356.62</td><td>326.50</td><td>276.12</td><td>170.15</td></tr><tr><td>Deviance</td><td>600.97</td><td>378.20</td><td>356.62</td><td>326.50</td><td>276.12</td><td>170.15</td></tr><tr><td>Pearson</td><td>442.88</td><td>439.37</td><td>424.69</td><td>1146.94</td><td>316.42</td><td>222.30</td></tr><tr><td>HL</td><td>5.41</td><td>2.15</td><td>7.55</td><td>34.19</td><td>2.21</td><td>3.66</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td>Best model</td></tr></table> Model and Results PREDICTION PERFORMANCE FOR LOGISTIC REGRESSION <table><tr><th></th><th>Model 1</th><th>Model 2</th><th>Model 3</th><th>Model 4</th><th>Model 5</th><th>Model 6</th></tr><tr><td>Type I error (%)</td><td>31.89</td><td>14.86</td><td>17.96</td><td>13.87</td><td>21.10</td><td>12.43</td></tr><tr><td>Type II error (%)</td><td>67.11</td><td>36.00</td><td>30.67</td><td>24.89</td><td>14.22</td><td>18.22</td></tr><tr><td>Average error (%)</td><td>33.48</td><td>15.80</td><td>18.53</td><td>14.36</td><td>20.79</td><td>12.69</td></tr><tr><td>Misclassification cost</td><td>110,405.20</td><td>91,170.89</td><td>91,451.59</td><td>85,212.72</td><td>87,064.02</td><td>80,377.25</td></tr></table> PREDICTION PERFORMANCE FOR DECISION TREE <table><tr><th></th><th>Model 1</th><th>Model 2</th><th>Model 3</th><th>Model 4</th><th>Model 5</th><th>Model 6</th><th>Model 7</th></tr><tr><td>Type I error (%)</td><td>34.10</td><td>26.99</td><td>25.02</td><td>19.61</td><td>18.32</td><td>16.89</td><td>15.86</td></tr><tr><td>Type II error (%)</td><td>61.78</td><td>20.89</td><td>27.56</td><td>12.00</td><td>23.56</td><td>17.78</td><td>20.44</td></tr><tr><td>Average error (%)</td><td>35.34</td><td>26.72</td><td>25.14</td><td>19.27</td><td>18.55</td><td>16.93</td><td>16.07</td></tr><tr><td>Misclassification cost</td><td>108,822.60</td><td>92,505.40</td><td>94,135.42</td><td>85,190.16</td><td>88,806.27</td><td>85,283.19</td><td>85,427.50</td></tr></table>					Goodness of fit	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	AIC	629.15	394.20	376.62	338.50	304.12	208.15	SC	641.54	427.23	417.91	363.28	361.92	286.61	-2 Log L	623.15	378.20	356.62	326.50	276.12	170.15	Deviance	600.97	378.20	356.62	326.50	276.12	170.15	Pearson	442.88	439.37	424.69	1146.94	316.42	222.30	HL	5.41	2.15	7.55	34.19	2.21	3.66							Best model		Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Type I error (%)	31.89	14.86	17.96	13.87	21.10	12.43	Type II error (%)	67.11	36.00	30.67	24.89	14.22	18.22	Average error (%)	33.48	15.80	18.53	14.36	20.79	12.69	Misclassification cost	110,405.20	91,170.89	91,451.59	85,212.72	87,064.02	80,377.25		Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Type I error (%)	34.10	26.99	25.02	19.61	18.32	16.89	15.86	Type II error (%)	61.78	20.89	27.56	12.00	23.56	17.78	20.44	Average error (%)	35.34	26.72	25.14	19.27	18.55	16.93	16.07	Misclassification cost	108,822.60	92,505.40	94,135.42	85,190.16	88,806.27	85,283.19	85,427.50
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Discussion The study faced imbalanced data proportion, thus it is required to perform data sampling in 1:1 proportion and extract balanced data samples. The study also faced the problem of multicollinearity for its large VIF variables. Future Work The future study will continue to perform data resampling, refine data collection process, and redesign new models to evaluate prediction performance in Logistic regression.		Conclusion In this study, data collection, and data preparation process such as data cleaning and data sampling, variables design amount to a long tedious process due to the huge data warehouse. Model 6 predicts well in both logistic regression and decision tree as observed in the prediction performance. Overall, we conclude that churn prediction model using logistic regression predicts better than decision tree model.																																																																																																																																							